

Reza Karimzadeh

PH.D., COMPUTER SCIENCE, COPENHAGEN UNIVERSITY

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SUMMARY

PhD candidate in Computer Science at the University of Copenhagen with hands-on experience in computer vision and deep learning across both academia and industry. I have built end-to-end machine learning systems for tasks such as object detection, 3D medical image segmentation, non-contact vital sign estimation, and explainable AI. Comfortable working from data preparation to model development and reproducible implementation, with experience leading small teams and collaborating across international and interdisciplinary environments. Seeking an industry role where strong ML engineering skills can be applied to real-world problems.

PROFESSIONAL EXPERIENCE

Visiting Researcher

UMC Amsterdam | VU Amsterdam, Netherlands

January 2026 - March 2026

- Treatment response prediction for Colorectal liver metastases (CRLM) patients

Computer Vision Developer

PART Artificial Intelligence Startup

October 2022 - January 2023

- Developed and deployed YOLO-based models for car plate detection and signature localization in official documents, improving detection accuracy and robustness in real-world applications

Research Collaborator

PINLab, Geneva University Hospital, Switzerland

January 2020 - September 2021

- Organs at risks and Tumor segmentation with deep learning methods

Head of Image Processing Group

Iran's National Elites Foundation

September 2020 - September 2021

- Led the Computer Vision team for non-contact Heart Rate estimation ([Link](#))
- Developed rPPG-based methods from face and palm RGB videos
- Designed data acquisition setups and preprocessing pipelines
- Coordinated integration with mobile and hardware-based solutions

Engineering Internship

Electro-Xray Company

July 2017 - September 2017

- Repair and maintenance of imaging systems (MRI, CT, Portable Radiology device, Angiography, C-Arm, Mammography and OPG)

Teaching Assistant

Copenhagen University | Sharif University of Technology | Amirkabir University of Technology

- Deep Learning Course | Numerical Methods
- Medical Images Analysis and Processing | Image Processing | Medical Imaging Systems

EDUCATION

🎓 Ph.D. Degree in Compute Science,

Copenhagen, Denmark

Image Section, DIKU, Copenhagen University

May 2023 - May 2026 (Expected)

Thesis: Applications of Deep Learning Algorithms for Medical Data Analysis

- A novel Explainable AI method based on Gradient-driven contours ([Link](#))
- Contrastive Learning Framework for 3D Patch-Based Medical Image Segmentation ([Link](#))
- Explanation-Guided of Missing Clinical Features Imputation for Survival Prediction
- Generating synthetic CT scans from MRI images for radiotherapy ([Link](#))
- Information Retrieval for radiological reports by exploiting public LLMs ([Link](#))

🎓 M.Sc. Degree in Biomedical Engineering,

Tehran, Iran

Sharif University of Technology

Sep. 2019 - May 2022

Thesis: Medical Image Segmentation Using Deep Learning Methods

- Attention-based deep learning segmentation: Application to brain tumor delineation ([Link](#))
- Proposing a shape-based loss function for organ segmentation in 3D medical images ([Link](#))
- COVID-19 detection from the chest computed tomography ([Link](#))

 B.Sc. Degree in Biomedical Engineering,

Tehran, Iran

Amirkabir University of Technology

Sep. 2014 - Sep. 2019

Thesis: Design and Implementation of Brain Surgery Bipolar Electrocautery Simulator Using Haptic Technology ([Link](#))

**TECHNICAL
SKILLS**

Programming

Python, MATLAB

Frameworks

PyTorch, MONAI, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib

CV & ML Methods

CNNs, Transformers, Segmentation, Object Detection, Self-Supervised Learning, XAI

MLOps / Systems

Git, GitHub, Docker, Linux

Documentation

L^AT_EX, Markdown, Microsoft Office

Languages

English, Persian, Danish (basic)

**OPEN SOURCE
PROJECTS**

Extremal Contours

Copenhagen University

[GitHub](#)

- Developed an algorithm-based tool for extracting and processing extremal image contours for robust object boundary analysis

Self-supervised Learning Framework for 3D Patch-based Image Segmentation

Copenhagen University

[GitHub](#)

- Implemented a self-supervised contrastive learning framework for 3D patch-based medical image segmentation with overlap-aware training

A Non-contact heart rate estimation framework

Iran's National Elites Foundation

[GitHub](#)

- Developed a non-contact heart rate estimation system using computer vision and remote PPG from facial video sequences
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**LEADERSHIP &
PROFESSIONAL
SERVICE**

Digital Infrastructure Coordinator (GitHub, Slack, Trello) & Symposium Organization

Image Section, DIKU, University of Copenhagen

Sep. 2023 - Present

- Managed shared research infrastructure and collaboration workflows
- Supported researcher onboarding and cross-project coordination
- Co-organized quarterly scientific symposia, coordinating speakers, scheduling, and session structure
- Facilitated interdisciplinary discussions and contributed to a strong internal research culture

Workshop Tutor/Organizer

Advanced Health Technologies-Artificial Intelligence in Medicine

December 2021

- Deep Learning workshop tutor and organizer ([Link](#))

Peer Review Service

Reviewer for international journals and conferences

2023 - Present

- Journal of Biomedical and Health Informatics | Journal of Medical Imaging | Scandinavian Conference on Image Analysis | NeurIPS'24 AIM-FM Workshop
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